Paragitja e sinjaleve

1° $x(t) = (1 - |t|)\underbrace{[u(t+1) - u(t-1)]}$

ndryshimja $(-1, 1)$

$t = -1$

$t = 0$

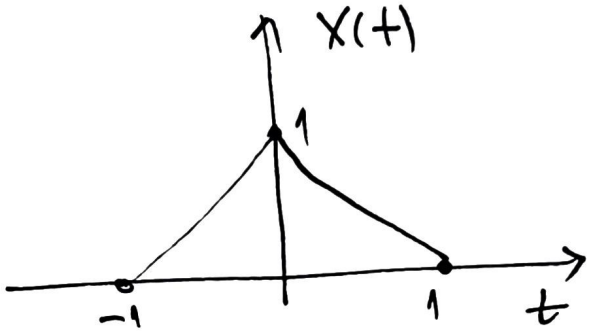
$t = 1$

$1 - |t|$

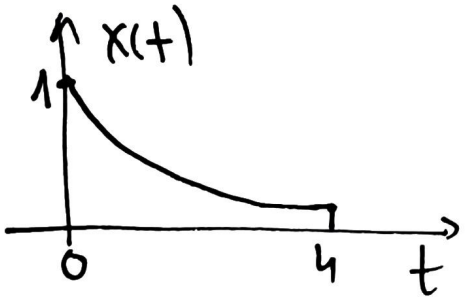
$1 - |0| = 1$

$1 - |-1| = 1 - 1 = 0$

$1 - |1| = 0$



2° $x(t) = e^{-t}(u(t) - u(t-4))$

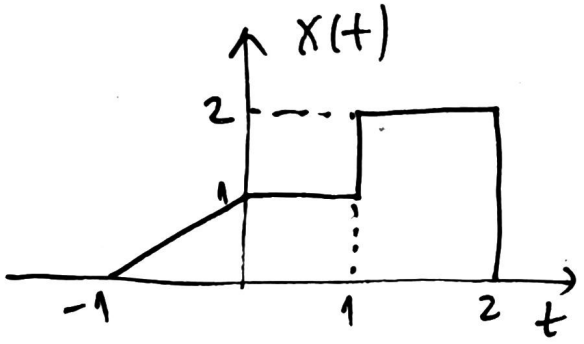


1.22 Sinjali kontinual $x(t)$ i dhënë në figurë. Skiconi sinjali në' vazhdim:

a) $x(t)\,u(1-t)$

b) $x(t)[u(t)-u(t-1)]$

c) $x(t)\delta\left(t-\frac{3}{2}\right)$



Zgjidhje

a)

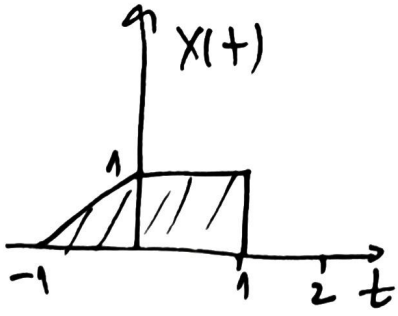
$$u(1-t)=\begin{cases}1\\0\end{cases}$$

$1-t\geq 0$

$-t\geq -1 \;/(-1)$

$t\leq 1$

$t>1$

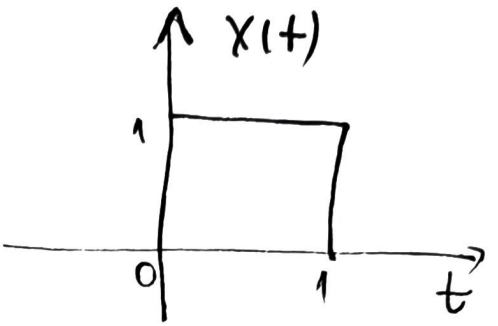


b) $u(t) - u(t-1) = \begin{cases} 1 & 0<t<1 \\ 0 & t>0 \end{cases}$

$t>0$ $\quad -t+1>0$

$-t>-1 \ (-1)$

$t<1$



c)

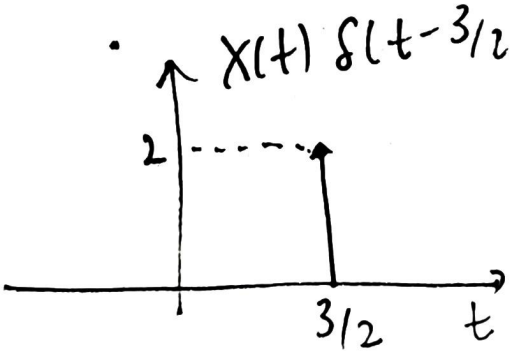
$$x(t)\,\delta\left(t-\frac{3}{2}\right) = x\left(\frac{3}{2}\right)\delta\left(t-\frac{3}{2}\right)$$

$t=\frac{3}{2}$

Vetia:

$$x(t)\,\delta(t-t_0) = x(t_0)\,\delta(t-t_0)$$

$t=t_0$



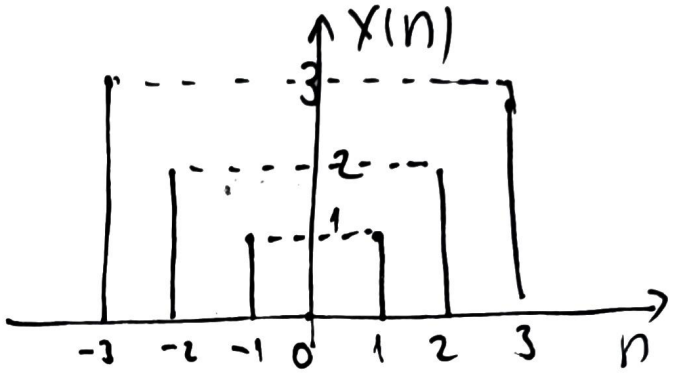
1.23

Eshtë dhënë sinjali diskret x(n) në figurë
Skiconi sinjalet në vazhdim:

a) $x(n)\,u(1-n)$

b) $x(n)\{u(n+2)-u(n)\}$

c) $x(n)\,\delta[n-1]$



a)

$$u(1-n)=\begin{cases}1\\0\end{cases}$$

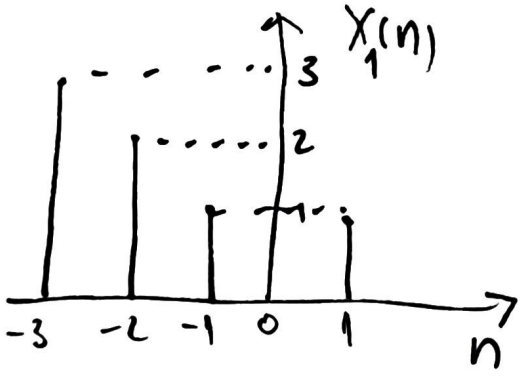
$1-n\geq 0$

$-n\geq -1 \;/(-1)$

$n\leq 1$

$\downarrow$

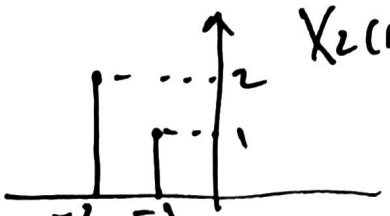
$n>0$



b) $u(n+2)-u(n)=\begin{cases}1\\0\end{cases}$

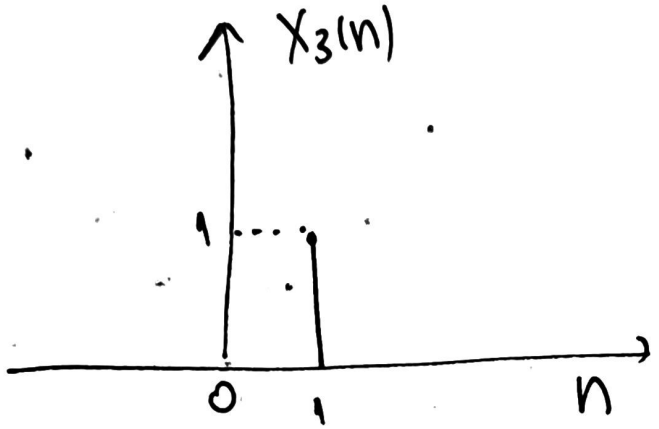
$-2\leq n<0$

$n>0$



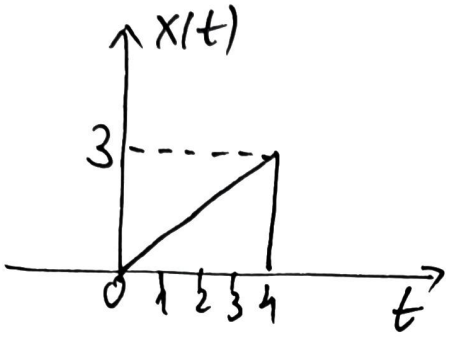
c) $x(n)\delta(n-1) = x(1)\delta(n-1) = \delta(n-1)$

$n=1$



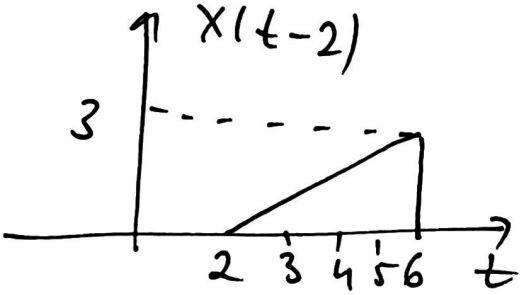
1.1 Për sinjalin kontinual $x(t)$ të paraqitur në figurë skiconi sinjalet në vazhdim:

a) $x(t-2)$ b) $x(2t)$ c) $x(t/2)$ d) $x(-t)$

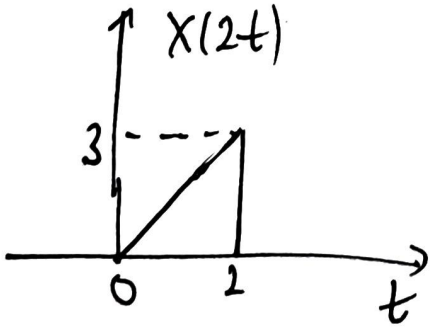


Zgjidhje

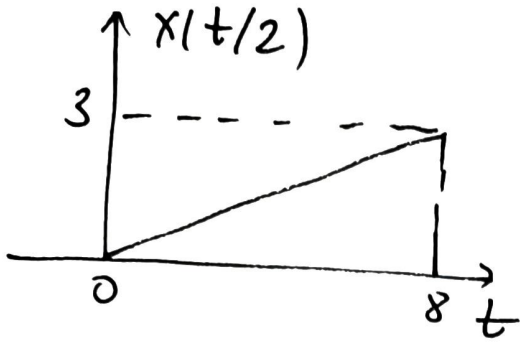
a) Zhvendosje për 2 njësi djathtas:



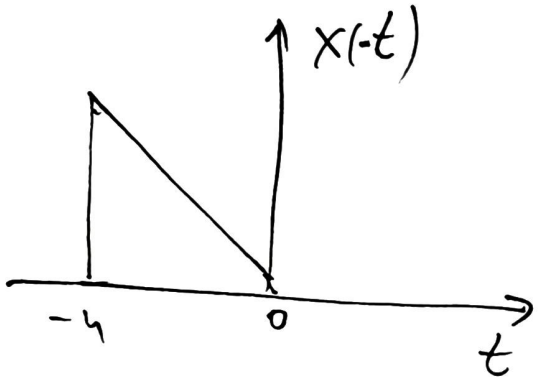
b) Ngushtim të brezit të sinjalit për 2 njësi



c) Zgjerim për 2 njësi

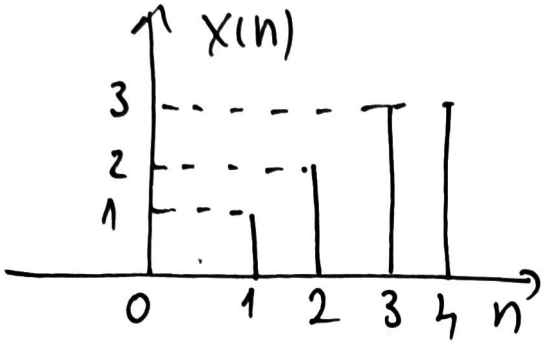


d) Pasqyrim

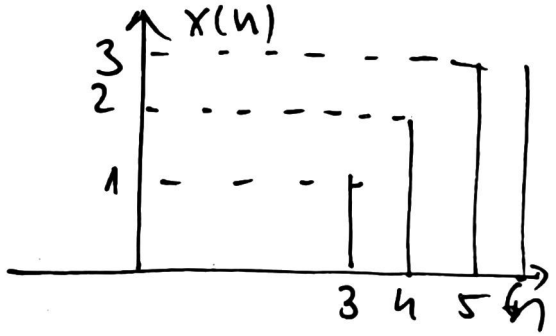


1.2 Për sinjalin diskret të dhënë në figurë skiconi sinjalet në vazhdim:

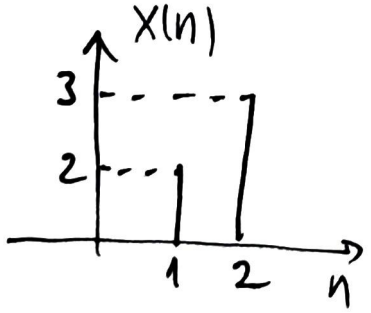
a) $x(n-2)$ b) $x(2n)$ c) $x(-n)$ d) $x(-n+2)$



a) Zhvendosje për 2 njësi djathtas:

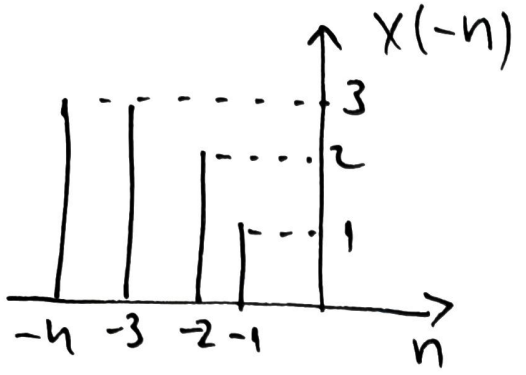


b) Ngushtim për 2 njësi

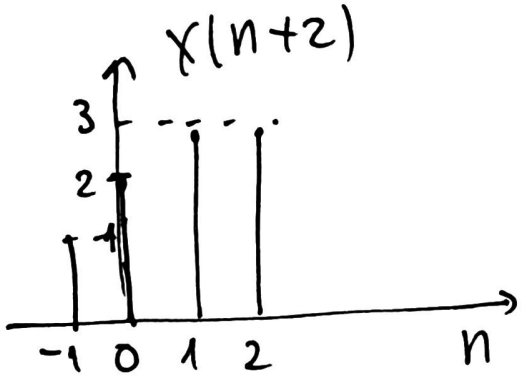


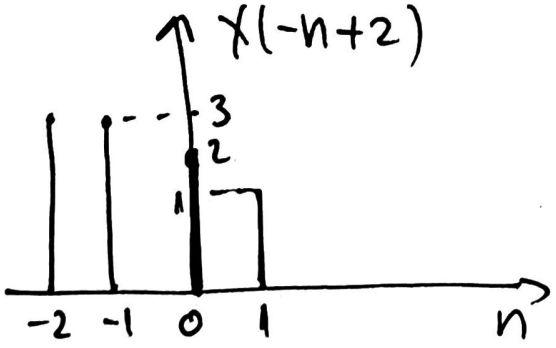
$\Rightarrow$ Ngushton vetem mostrat që dalin me vlera të plota!

c) Pasyginim



d) Kemi dy veti — Zhvendosje majtas për 2
Pasyginim





1.7

$$x(n) = \underbrace{(6-n)}_{\downarrow}\underbrace{[u(n) - u(n-6)]}_{\downarrow}$$

f-oni shtrinja

a) $y_1(n) = x(2n-3)$

b) $y_2(n) = x(8-3n)$

Zgjidhje

shtrinja prej (0,5)

n=0
n=1
n=2
n=3
n=4
n=5

} Zëvendesojm tek f-oni (6-n)

